

Name: I Forgot Team Date: _____

Class: _____

MyDesign Portfolio

Element A Initial Template

Element A: Presentation and Justification of the Problem

Problem Statement and Details (A1, A2, A5)

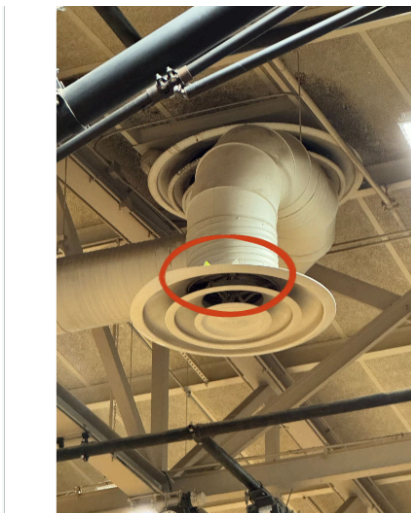
Ms. Hopkins needs to get the badminton birdies off of the gym second floor window ledges because it costs the school resources and money to fix and means the badminton players can't play. The fact that they can't play when this situation happens is concerning, since physical education provides cognitive content and instruction designed to develop motor skills, knowledge and behavior for physical activity and physical fitness. Supporting schools to establish physical education daily can provide students with the ability and confidence to be physically active for a lifetime (CDC 2022). The school, Ms. Hopkins, and the badminton players all need us to innovate on Ms. Hopkins' tool for getting badminton birdies off of the roof. This affects the community because the badminton players can't play, Ms. Hopkins can't teach, and the school has to pay money for new equipment and fixes. For example, every time Ms. Hopkins runs out of birdies, the school either buys expensive new equipment, which costs around 20\$ to 25\$, and considering that they would have to spend this money probably every month, it's a lot, or pays building services to clean off the roofs whenever they change the lights in the gym, which over time adds up and takes up a large part of the budget, which limits other equipment for the gym classes. Our job is to create a solution to the problem to clean the gym roof of badminton birdies.

Due to the height and shape of the gym and the birdies getting stuck in many places, such as the basketball hoops, rafters, and windows, this situation occurs all through the gym.

The problem affects the people who work and study at Winston Churchill High School, mostly the badminton and tennis players. It also affects the gym teachers who need to spend their money on new birdies every year. The problem is physically happening at the Winston Churchill gymnasium and it has been happening for years. It happens because the

Name: I Forgot Team Date: _____ Class: _____

balls are hit with too much force or at an angle that causes them to go higher than expected and it happens very often.



Images by: Miss Hopkins (stakeholder)

Stakeholder Concerns and Connections (A3, A4)

Stakeholder 1: The badminton players are important stakeholders because they lose their birdies constantly and aren't able to keep playing.

Name: I Forgot Team **Date:** _____ **Class:** _____

Stakeholder 2: Ms. Hopkins is the next stakeholder, who has to pay for the new equipment and it disturbs her class because of the missing birdies. She also created the first design to try to fix the problem, but it failed due to the durability.

Stakeholder 3: The school is a valuable stakeholder because they have to pay for any equipment. Also, missing birdies take away from the budget from the PE department, meaning they have less equipment for other learning.

Stakeholder 4: The PE students lose their birdies and are greatly effected by the problem as they can't learn about badminton without the birdies.

Stakeholder 5: The Engineering teacher is the last stakeholder as they are incharge of organizing possible solutions to the badminton birdie problem.

Concerns of the stakeholders:

Some of the biggest concerns of the stakeholders are losing their equipment because they can't reach the tall places where the balls get stuck, and also the money that they have to spend every year with more equipment. To save their money and to avoid losing so many new and in perfect condition materials we will help solve this problem. This combines all of the stakeholders to try to come up with solutions through the engineering students.

Where is this a problem?

This problem can be happening at other schools too, so solving this problem for Winston Churchill High School, might also help other schools who share the same problem. Also, if something gets stuck in a tree, then we can use the solution to get the thing down.

Problem Importance (A6)

It's important to solve the problem of sports balls and other materials getting stuck in high places at a high school gymnasium for a few reasons. First, it can be unsafe if someone tries to retrieve them without proper equipment or training. Second, it can disrupt the flow of the game or activity if players have to pause to retrieve the stuck items. Lastly, it can be costly to replace lost or damaged equipment. So, it's best to address the issue promptly to ensure a safe and smooth sporting experience.

References List at the end of this document

Ms. Hopkins

Name: I Forgot Team Date: _____

Class: _____

Engineering Teacher

"Physical Education." Edited by CDC.com, Centers for Disease Control and Prevention, Centers for Disease Control and Prevention, 26 July 2022, www.cdc.gov/healthyschools/physicalactivity/physical-education.htm#:~:text=Physical%20education%20provides%20cognitive%20content,physical%20activity%20and%20physical%20fitness.&text=Supporting%20schools%20to%20establish%20physical,physically%20active%20for%20a%20lifetime.

Element A: Presentation and Justification of the Problem						
A1. Problem identification is _____.	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
	very clear and objective	clear and objective	somewhat clear and objective	somewhat clear or objective	unclear	missing (cannot be inferred from information included)
A2. Problem definition is _____.	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
	in considerable depth	in more than adequate depth	in adequate depth	in superficial depth	unclear	missing (cannot be inferred from information included)
A3. Problem elaboration is _____.	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
	well elaborated	elaborated	imprecisely or generally elaborated	minimally elaborated	unelaborated and/or is clearly subjective	not elaborated (cannot be inferred from information included)
A4. Concern of primary	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0

Name: I Forgot Team Date: _____ Class: _____

stakeholder groups. Justification of the problem highlights the concerns of _____.	almost all primary stakeholder groups	many primary stakeholder groups	several primary stakeholder groups	at least a few primary stakeholders /stakeholder groups	only one or two stakeholders /stakeholder groups	no primary stakeholders /stakeholder groups
A5. Problem justification is based on _____.	score = 5 numerous, timely, credible, and various sources	score = 4 timely, credible, and various sources	score = 3 credible and timely sources	score = 2 credible sources	score = 1 partially-credible sources	score = 0 essentially only the opinion of the researcher
A6. Problem justification offers _____-objective detail.	score = 5 consistently	score = 4 mostly	score = 3 generally	score = 2 somewhat	score = 1 unclear	score = 0 no